

Math 170S - Homework 2 - due January 22

- Numbers refer to the exercise numbers in the Hogg, Tanis, Zimmerman textbook.
- Unless specified otherwise, do all the subparts of a problem.
- If there are additional questions after the number of the problem, you should answer them too.
- Show all your work.

Problem 1. 6.1 - 5

Problem 2. 6.2 - 6. You do not have to find and draw fences (but you can read about them in Section 6.2 and do it:))

Problem 3. 6.3 - 5

Problem 4. 6.3 - 14

Problem 5. 6.3 - 3 b) and c) only. Do not use the formula for PDF of Y_i from the textbook, but rather derive the probabilities in question from the meaning of order statistics.

Hint: If $Y_4 < 5$, then how many of the samples can be greater than 5? What is the probability that exactly one sample is greater than 5? See also Example 6.3-2.

Problem 6. 6.4 - 1 and 6.4 - 2.

What is similar and what is crucially different comparing to the example that we did in class, when we estimated two unknown parameters of normal distribution $N(\theta_1, \theta_2)$?

Problem 7. 6.4 - 7

Problem 8. Let $X_1, X_2, \dots, X_n$ be independent samples taken from some distribution, and $\bar{X}$ is the sample mean. Let random variable X have the same distribution. Prove that

$$a) \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \right] = \mathbb{E} X^2 - \mathbb{E}(\bar{X})^2.$$

$$b) \sum_{i=1}^n (X_i - \mathbb{E} X)^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mathbb{E} \bar{X})^2.$$